# Meet the Microbes

*What are bacteria, and how are they different from our cells?*

Remember: All living things are made of cells. Some living things are made of just one cell. Bacteria are one of them! Today we'll compare a bacterium to the cells we studied last unit.

## PART A — NASA News: Bacteria That Plays Dead

> ***Read (or listen to your teacher read):*** *In 2007, engineers building NASA's Phoenix Mars lander did it inside one of the cleanest rooms on Earth — blasted with UV lights and chemicals to wipe out every living thing. They thought it worked. They were wrong. Scientists later found a brand-new bacterium hiding there: Tersicoccus phoenicis. Even stranger, the same species turned up in a second clean room 2,500 miles away. Nobody could explain how it survived — until 2025. Researchers discovered it can shut its entire metabolism down, entering a state so deep that standard tests read it as dead. "It is not dead. It was playing dead," said microbiologist Madhan Tirumalai. "It is only dormant." Even more unsettling: the Phoenix lander was launched to Mars in 2007 — meaning we may have accidentally sent bacteria to the planet we were searching for alien life.*

*Source: Robin George Andrews, National Geographic, October 2025 · nationalgeographic.com/science/article/clean-room-bacteria-nasa-spacecraft-dormant*

**1. What did scientists discover about *Tersicoccus phoenicis* in 2025?**

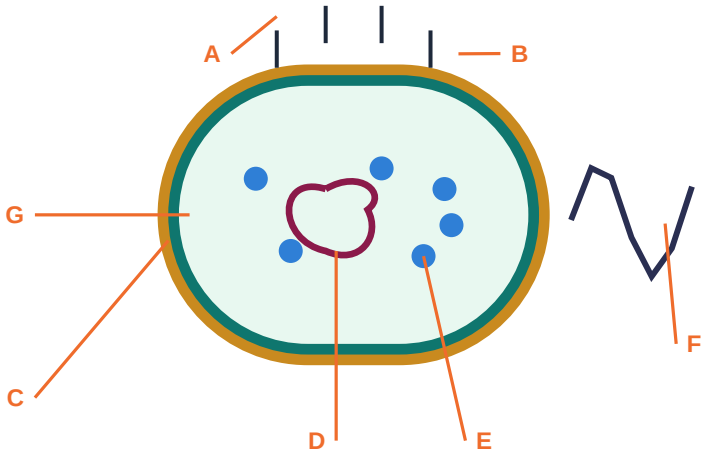
**2. The article says the bacteria "plays dead." What do you think the word dormant means?**

**3. Why would NASA care about bacteria surviving on a spacecraft? (Hint: think about what NASA was looking for on Mars.)**

## PART B — Label the Bacterium

Use the word bank. Write the correct part name next to each letter in the table.

| cell wall | cell membrane | cytoplasm | DNA | ribosomes | flagellum | pili |
|---|---|---|---|---|---|---|




| Letter | Part |
|---|---|
| A | |
| B | |
| C | |
| D | |

| Letter | Part |
|---|---|
| E | |
| F | |
| G | |

*A bacterium (prokaryote). Real size: ~1–2 micrometers — 500 could line up across the head of a pin.*

## PART C — Compare: Bacterium vs. Our Cells

Write **Yes** or **No** in each box.

| Cell part | Bacterium | Animal cell (yours) | Plant cell |
|---|---|---|---|
| Nucleus | | | |
| Cell membrane | | | |
| Cell wall | | | |
| Ribosomes | | | |
| Mitochondria | | | |
| Chloroplasts | | | |
| DNA | | | |
| Flagellum (tail) | | | |

**4. Which cell from our last unit is a bacterium MOST like? Explain using the table.**

## PART D — Connect to Cell Transport

A bacterium's cell wall keeps it from bursting. Inside the cell there is a LOT of dissolved stuff, so water constantly tries to move IN by osmosis. The wall pushes back like the tough outer shell of a soccer ball holding in the air.

**5. The antibiotic penicillin breaks down the cell wall. Predict what happens to the bacterium next. Use the word osmosis.**

**6. Penicillin does not hurt YOUR cells this way. Use the table to explain why.**

**7. Bacteria don't have mouths. How do you think food molecules get into a bacterium? (Think: diffusion, transport proteins, active transport.)**

**8. Tie it to the NASA story: Scientists can "wake up" dormant *T. phoenicis* using a protein — and once awake, antibiotics can kill it. Why can't antibiotics kill it while it's dormant? Use what you know about how bacteria survive.**

## PART E — Bacteria Math: Binary Fission

Bacteria reproduce by **binary fission**: one cell splits into two. Under perfect conditions, some bacteria split every 20 minutes. Start with ONE bacterium on your water bottle.

| Time | 0 min | 20 min | 40 min | 1 hour | 2 hours | 4 hours | 6 hours | 8 hours |
|---|---|---|---|---|---|---|---|---|
| Bacteria | 1 | 2 | | | | | | |

**9. Why does this fast reproduction matter when we're trying to fight an infection?**

## PART F — Helpful, Harmful, or Neutral?

Most bacteria are NOT germs. Sort each one below. Write **Helpful**, **Harmful**, or **Neutral**.

**Lactobacillus**
Turns milk into yogurt.
→ _____

**Gut E. coli**
Lives in your intestines and makes vitamin K.
→ _____

**Streptococcus**
Causes strep throat.
→ _____

**Vibrio cholerae**
Spreads in dirty water and causes cholera.
→ _____

**Rhizobium**
Lives on plant roots and makes nitrogen fertilizer.
→ _____

**MRSA**
A staph "superbug" that can infect turf burns.
→ _____

**Soil decomposers**
Break down dead leaves and recycle nutrients.
→ _____

**Skin bacteria**
Live on your skin and crowd out germs.
→ _____

**Ocean bacteria**
Float in seawater, using sunlight to grow. Most never touch us.
→ _____

■ **Exit Ticket** Name **2 ways** a bacterium is like your cells and **1 way** it is different.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. NASA news: National Geographic, October 2025. Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html